

Week 1 Task DevOps intern

Linux Fundamentals and Basic Administration

Introduction

Linux is an open-source operating system widely used in servers, cloud platforms, and DevOps environments. Most web servers, containers, and cloud infrastructure run on Linux because it is stable, secure, lightweight, and highly customizable. Understanding Linux fundamentals is essential for DevOps engineers because they frequently work with servers, automation tools, containers, and cloud environments.

1. Linux Fundamentals

Linux is an operating system based on the Linux kernel. It manages hardware resources, provides a user interface, and allows users and applications to interact with the system.

A Linux system mainly consists of:

- **Kernel** – The core part of Linux that manages CPU, memory, devices, and processes.
- **Shell** – A command-line interface that allows users to interact with the operating system.
- **File System** – A method of organizing and storing files.
- **Applications and Services** – Programs that run on the system.

Common Linux distributions include:

- Ubuntu
- Debian
- Fedora
- CentOS
- Red Hat Enterprise Linux

In DevOps, Linux is commonly used for hosting applications, managing servers, running containers, and automating deployments.

2. Linux File System

Linux uses a hierarchical file system structure where everything starts from the root directory (/).

Important Linux directories:

Directory	Purpose
/	Root directory, starting point of the filesystem
/home	Contains user home directories
/etc	Contains system configuration files
/var	Stores logs and variable data
/bin	Contains essential commands

Directory	Purpose
/usr	Contains user applications and utilities
/tmp	Temporary files
/root	Home directory of the root user
/dev	Contains device files

Unlike Windows, Linux does not use drive letters like C : or D : . All storage devices are mounted under the root directory.

3. Users and Groups

Linux is a multi-user operating system where multiple users can access the same system. Users and groups help manage access and security.

Users

A user account represents an individual or service that can access the system.

Create a user:

```
sudo adduser username
```

or:

```
sudo useradd username
```

Groups

Groups allow multiple users to share permissions.

Create a group:

```
sudo groupadd developers
```

Add a user to a group:

```
sudo usermod -aG developers username
```

Linux also uses a root user, which has complete administrative privileges.

4. File Permissions

Linux uses file permissions to control who can read, modify, or execute files.

Every file has three permission categories:

- **Owner** – The user who owns the file
- **Group** – Users belonging to the file group
- **Others** – All other users

Permissions:

Permission Meaning

r	Read
w	Write

Permission Meaning

x Execute

Example:

```
ls -l file.txt
```

Output:

```
-rwxr-xr-- file.txt
```

Meaning:

- Owner: read, write, execute
- Group: read, execute
- Others: read

Changing permissions:

```
chmod 755 file.sh
```

Changing ownership:

```
sudo chown user:group file.txt
```

File permissions are important in DevOps because they protect configuration files, scripts, and sensitive data.

5. Package Management

Package management allows users to install, update, and remove software easily.

Ubuntu and Debian-based systems use **APT**.

Update package list:

```
sudo apt update
```

Install software:

```
sudo apt install package_name
```

Remove software:

```
sudo apt remove package_name
```

Upgrade installed packages:

```
sudo apt upgrade
```

6. Environment Variables

Environment variables store system and user-specific configuration values.

They are used by applications and scripts to access important settings.

View environment variables:

`printenv`

View a specific variable:

`echo $PATH`

Common environment variables:

Variable	Purpose
PATH	Locations where commands are searched
HOME	User home directory
USER	Current username
SHELL	Current shell

Create a variable:

`export NAME="DevOps"`

7. Basic Networking

Networking allows Linux systems to communicate with other machines.

Important networking concepts:

- IP Address – Identifies a device on a network
- DNS – Converts domain names into IP addresses
- Port – Identifies network services
- Protocol – Rules for communication (TCP, UDP, HTTP, SSH)

Useful commands:

Check IP address:

`ip addr`

Check connectivity:

`ping google.com`

Check open ports:

`ss -tuln`

DNS lookup:

`nslookup google.com`

8. SSH (Secure Shell)

SSH is a secure protocol used to remotely access and manage Linux servers.

Basic syntax:

`ssh username@ip_address`

Example:

```
ssh user@192.168.1.10
```

SSH provides encrypted communication and is commonly used by DevOps engineers to manage remote servers.

Copy files using SCP:

```
scp file.txt user@server:/home/user/
```

SSH keys are often used instead of passwords for secure authentication.

9. 20 Common Linux Commands

Command	Purpose
pwd	Shows current directory
ls	Lists files and directories
cd	Changes directory
mkdir	Creates a directory
rmdir	Removes empty directory
touch	Creates a file
cp	Copies files
mv	Moves or renames files
rm	Removes files
cat	Displays file contents
less	Views large files
head	Shows beginning of file
tail	Shows end of file
grep	Searches text
find	Searches files
nano	Text editor
vim	Advanced text editor
ps	Shows running processes
top	Monitors system resources
df	Shows disk usage

Example:

```
ls -la
```

Displays all files including hidden files with detailed information.